

Complete Machine Learning Notes (Hinglish)

Topics Covered

1. Data Preprocessing
 2. Handling Missing Values
 3. Encoding Categorical Variables
 4. Cleaning and Transforming Dataset
 5. KNN Classification Algorithm
 6. Regression Algorithms in Python
 7. Spam Detection using SVM
 8. Performance Metrics
 9. Confusion Matrix
 10. Data Visualization Techniques
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1. Data Preprocessing

Definition

Data preprocessing ka matlab raw data ko clean aur proper format me convert karna hota hai taaki Machine Learning model usko easily samajh sake.

Real-world data generally:

- incomplete,
- noisy,
- duplicate,
- inconsistent

hota hai.

Importance of Data Preprocessing

1. Accuracy Improve karta hai

Clean data se model better learning karta hai.

2. Missing Values Handle karta hai

NULL values ko remove ya fill karta hai.

3. Noise Remove karta hai

Wrong values aur duplicates remove karta hai.

4. Machine Readable banata hai

Text ko numbers me convert karta hai.

5. Feature Scaling karta hai

Large values ko balance karta hai.

Main Steps

1. Data Cleaning
 2. Data Integration
 3. Data Transformation
 4. Data Reduction
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2. Handling Missing Values

Definition

Dataset me missing values ko remove ya replace karne ki process ko handling missing values bolte hain.

Methods

1. Remove Rows

Missing row delete.

2. Remove Columns

Zyada missing values ho to column delete.

3. Fill with Mean

Numerical data ke liye.

4. Fill with Median

Outliers ho to useful.

5. Fill with Mode

Categorical data ke liye.

Example

Name	Age
A	20
B	NULL

3. Encoding Categorical Variables

Definition

Text/categorical data ko numbers me convert karna encoding kehlata hai.

Types of Encoding

1. Label Encoding

Example:

- Red → 0
 - Blue → 1
-

2. One Hot Encoding

Har category ke liye separate column.

Red	Blue	Green
1	0	0

Importance

- Machine readable data
 - Better training
 - Improved accuracy
-

4. Cleaning and Transforming Dataset

Incomplete Dataset

Jisme missing values ho.

Noisy Dataset

Jisme errors/outliers ho.

Data Cleaning

Includes:

- missing values handling
 - duplicate removal
 - error correction
 - noise removal
-

Data Transformation

Includes:

- normalization
 - standardization
 - encoding
 - feature scaling
-

Example

Raw data:

- duplicates
- missing values
- wrong entries

After preprocessing:

- clean dataset
 - better ML performance
-

5. KNN (K-Nearest Neighbors)

Definition

KNN ek supervised learning classification algorithm hai jo nearest neighbors ki help se prediction karta hai.

Working of KNN

Step 1: K choose karo

Step 2: Distance calculate karo

Euclidean Distance Formula:

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Step 3: Nearest neighbors find karo

Step 4: Majority voting

Most repeated class final prediction.

Advantages

- Simple

- Easy to understand
 - No training phase
-

Disadvantages

- Large data me slow
 - Scaling required
-

Python Code

```
from sklearn.neighbors import KNeighborsClassifier

model = KNeighborsClassifier(n_neighbors=3)
```

Applications

- Recommendation systems
 - Image recognition
 - Medical diagnosis
-

6. Regression Algorithms in Python

Regression Kya Hota Hai?

Regression numerical values predict karta hai.

Examples:

- house price prediction
 - salary prediction
 - weather prediction
-

Types of Regression

1. Linear Regression

Straight line relationship ko represent karta hai.

Equation

$$y = mx + c$$

Where:

- y = output
 - x = input
 - m = slope
 - c = intercept
-

Example

Experience badhega $\rightarrow$ Salary badhegi.

Python Code

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
```

2. Polynomial Regression

Jab relationship straight line me na ho aur curved ho tab polynomial regression use hota hai.

Example

- Population growth
 - Stock trends
 - Temperature changes
-

Polynomial Equation

$$y = a + bx + cx^2$$

Python Code

```
from sklearn.preprocessing import PolynomialFeatures
```

3. Logistic Regression

Naam regression hai but mainly classification ke liye use hota hai.

Output:

- Yes/No
- True/False
- Spam/Not Spam

Example

- Email spam detection
- Disease prediction
- Pass/Fail prediction

Sigmoid Function

Logistic regression sigmoid curve use karta hai.

Output Range

0 se 1 ke beech probability.

Python Code

```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression()
```

Working of Regression

1. Dataset lo
 2. Best fit line/curve find karo
 3. Prediction karo
-

Cost Function

MSE Formula:

$$MSE = (1/n) \sum (y - \hat{y})^2$$

Applications

- Stock prediction
 - Sales prediction
 - House price prediction
 - Spam detection
 - Medical prediction
-

7. Spam Detection using SVM

SVM Kya Hota Hai?

Support Vector Machine ek supervised classification algorithm hai.

Spam detection me use hota hai.

Main Idea

SVM best separating boundary find karta hai.

Is boundary ko hyperplane bolte hain.

Important Terms

1. Hyperplane

Decision boundary.

2. Support Vectors

Nearest important points.

3. Margin

Boundary aur nearest points ka distance.

Steps in Spam Detection

1. Dataset collect
 2. Text preprocessing
 3. TF-IDF conversion
 4. Train SVM
 5. Prediction
-

Python Code

```
from sklearn.svm import SVC  
  
model = SVC(kernel='linear')
```

Applications

- Spam detection
 - Face recognition
 - Sentiment analysis
-

8. Performance Metrics

Definition

Performance metrics ML model ki performance evaluate karte hain.

Common Metrics

1. Accuracy
 2. Precision
 3. Recall
 4. F1-score
-

Accuracy

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Precision

$$\text{Precision} = TP / (TP + FP)$$

Recall

$$\text{Recall} = TP / (TP + FN)$$

F1-Score

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

9. Confusion Matrix

Definition

Confusion matrix actual aur predicted values compare karti hai.

Structure

Actual / Predicted	Positive	Negative
Positive	TP	FN
Negative	FP	TN

Terms

TP

True Positive

TN

True Negative

FP

False Positive

FN

False Negative

Example

	Predicted Spam	Predicted Normal
Actual Spam	40	10
Actual Normal	5	45

10. Data Visualization Techniques

Definition

Data ko graphical form me represent karna data visualization kehlata hai.

Popular Libraries

1. Matplotlib
 2. Seaborn
 3. Plotly
-

Types of Visualization

1. Line Chart

Trend show karta hai.

```
plt.plot(x,y)
```

2. Bar Chart

Comparison ke liye.

```
plt.bar(x,y)
```

3. Histogram

Distribution show karta hai.

```
plt.hist(data)
```

4. Scatter Plot

Relationship between variables.

```
plt.scatter(x,y)
```

5. Pie Chart

Percentage distribution.

```
plt.pie(data)
```

Importance of Visualization

- Easy understanding
 - Pattern detection
 - Better decision making
 - Outlier identification
-

Short Exam Definitions

Data Preprocessing

Raw data ko clean aur transform karna.

KNN

Nearest neighbors ke basis par classification.

Regression

Continuous numerical values prediction.

SVM

Classification algorithm using hyperplane.

Confusion Matrix

Actual aur predicted values comparison table.

Data Visualization

Charts aur graphs ke through data representation.